Derivation

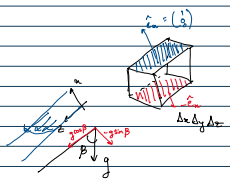
Thursday, 21 September 2023 6:32 PM

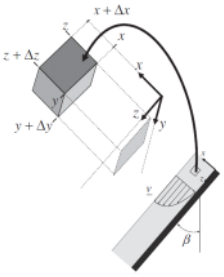
Inclined plane Derivation

$$f = \int\int_A (\hat{n}\cdot\underline{\underline{\Pi}})\,dS$$

$$(\hat{n}\cdot\underline{\underline{\Pi}})\Big|_{top} = \underset{1\times3}{(1\;0\;0)}\begin{pmatrix} \tau_{xx}-p & \tau_{xy} & \tau_{xz} \\ \vdots & \ddots & \vdots \\ \cdots & \cdots & \cdots \end{pmatrix}_{3\times3}$$

$$= (\tau_{xx}-p \quad \tau_{xy} \quad \tau_{xz})$$





$$\begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix}_{3\times1}\underset{1\times3}{(v_x\;\;v_y\;\;v_z)} = \begin{pmatrix} \frac{\partial v_x}{\partial x} & \frac{\partial v_y}{\partial x} & \frac{\partial v_z}{\partial x} \\ \frac{\partial v_x}{\partial y} & \frac{\partial v_y}{\partial y} & \frac{\partial v_z}{\partial y} \\ \frac{\partial v_x}{\partial z} & \frac{\partial v_y}{\partial z} & \frac{\partial v_z}{\partial z} \end{pmatrix}$$

$$\sum_{on\ CV} f = 0 = f_{gravity} + \sum_{i=1}^{6} f_{i^{th}\ side}$$

↳ force balance on a fluid element

$$\sum f = \begin{pmatrix} -\rho g\sin\beta \\ 0 \\ \rho g\cos\beta \end{pmatrix}\Delta x\Delta y\Delta z + \left[\Delta y\Delta z\begin{pmatrix} \tau_{xx}-p \\ \tau_{xy} \\ \tau_{xz} \end{pmatrix}_{xyz}\Big|_{x+\Delta x} - \Delta y\Delta z\begin{pmatrix} \tau_{xx}-p \\ \tau_{xy} \\ \tau_{xz} \end{pmatrix}_{xyz}\Big|_{x} \cdots\cdots\right]$$

↳ Split into each x, y, z component & taking limits

$$x: \quad 0 = -\rho g\sin\beta + \left(\frac{\tau_{xx}|_{x+\Delta x} - \tau_{xx}|_x}{\Delta x}\right) - \left(\frac{p|_{x+\Delta x} - p}{\Delta x}\right) + \left(\frac{\tau_{yx}|_{y+\Delta y} - \tau_{yx}|_y}{\Delta y}\right) + \left(\frac{\tau_{zx}|_{z+\Delta z} - \tau_{zx}|_z}{\Delta z}\right)$$

↳ Limit:

$$x: \quad \boxed{0 = -\frac{\partial p}{\partial x} + \left[\frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z}\right] - \rho g\sin\beta}$$

x component momentum balance!

[4.357, 4.358, 4.359]

↓

But our goal is to find velocity profile!

$$\tau = \mu\left(\nabla\underline{v} + (\nabla\underline{v})^T\right)$$

↳ Need to use an equation which now relates τ & v → Newtonian constitutive equation

$$v = \begin{pmatrix} 0 \\ 0 \\ v_z(x) \end{pmatrix} \quad ; \quad \underline{\tau} = \mu\begin{pmatrix} 2\frac{\partial v_x}{\partial x} & \frac{\partial v_y}{\partial x}+\frac{\partial v_x}{\partial y} & \frac{\partial v_z}{\partial x}+\frac{\partial v_x}{\partial z} \\ \frac{\partial v_x}{\partial y}+\frac{\partial v_y}{\partial x} & 2\frac{\partial v_y}{\partial y} & \frac{\partial v_z}{\partial y}+\frac{\partial v_y}{\partial z} \\ \frac{\partial v_x}{\partial z}+\frac{\partial v_z}{\partial x} & \frac{\partial v_y}{\partial z} & 2\frac{\partial v_z}{\partial z} \end{pmatrix}_{xyz}$$

$$\tau = \mu\begin{pmatrix} 0 & 0 & \frac{\partial v_z}{\partial x} \\ 0 & 0 & 0 \\ \frac{\partial v_x}{\partial x} & 0 & 0 \end{pmatrix}_{xyz}$$

(τ_{xx} at top-left, τ_{zy} at bottom-middle)

$$\tau_{xz} = \mu\frac{dv_z}{dx}$$

$$\tau_{zx} = \mu\frac{dv_z}{dx}$$

↓

Substitute in balance to arrive at simplified form

There will be only 1 direction where velocity would remain for unidirectional flows!

$$z: \quad \boxed{0 = -\frac{\partial p}{\partial z} + \mu\frac{d^2v_z}{dx^2} + \rho g\cos\beta}$$

↓

Solve for v_z w/ BC's:

@ $x = 0$

@ $x = H$; $\tau_{xz} = 0 = \mu\frac{dv_z}{dx}$

$$\boxed{v_z(x) = \frac{\rho g\cos\beta}{\mu}\left[Hx - \frac{x^2}{2}\right]}$$

Pg 380, 381, 382